WAREHOUSE MANAGEMENT SYSTEM

Submitted By:

- **PULKIT MISHRA, 19**
- **AKHIL TYAGI, 03**

5/18/2011

ACKNOWLEDGEMENT

We sincerely thank and acknowledge the co-operation and guidance provided by the department, faculties involved during the completion of entire project.

We are also grateful to our faculty, Mr Abhinav Garg for guiding us in every possible way and providing us with the constant support and encouragement to complete our RDBMS project.

Introduction

A Warehouse Management System or WMS, is the key part of the supply chain, and primarily aims to control the movement and storage of materials within the warehouse and process the associated transactions including shipping, receiving, picking and putaway.

The system also directs and optimizes stock putaway based on the real time information about the status of bin utilization/space management.

Warehouse management is also the detailed co-ordination of a complex operation involving many people, facilities, and supplies.

Need of the System

The need of a warehouse system is to provide set of computerized procedures to handle the receipt of stock and returns into a warehouse facility, model and manage the logical representation of the physical storage facility (e.g., racking etc.), manage the stock within the facility and enable a seamless link to order processing and logistics management in order to pick, pack and ship product out of the facility.

The primary purpose of a WMS is to control the movement and storage of materials within a warehouse – you might even describe it as the lags at the end – of – line which automates the store, traffic and shipping management.

Scope of the System

In its simplest form, the WMS can data track products during the production process and act as an interpreter and message buffer between existing ERP or a common database that supports all application across an entire organization and warehouse management system.

Warehouse Management is not just managing within the boundaries of a warehouse today; it is much wider and goes beyond the physical boundaries. Inventory management, inventory planning, cost management, IT applications and communication technology to be used are all related to warehouse management. The container storage, loading and unloading are also covered by warehouse management today. Warehouse management today is part of supply chain management and demand management. Even production management is to a great extent dependent on warehouse management.

Efficient Warehouse Management gives a cutting edge to a retail chain distribution company.

Warehouse management does not just start with receipt of material but it actually starts with actual initial planning when container design is made for a product.

The detailed scope of a warehouse management includes:-

- It includes a record of the items that are in the warehouse or existing stock.
- The scope extends to efficient inclusion and deletion of the variety in goods.
- The system includes management of the precise position of the goods within the warehouse.

- Scope includes the duration of time the pick-up or putaway of the stock.
- Also, making note on a detailed level keeping track of the date of order, date of dispersal of goods and date of retrieval, thereby enabling further analysis and modification in the design, if needed.

System Planning

Warehouse Management monitors the progress of products through the Warehouse. It involves the physical warehouse infrastructure, tracking system and communication between product stations.

Warehouse Management System uses the process of Wave Picking to support managing the work of a warehouse or distribution centre.

There are two basic planning elements and benefits of Wave Picking

1. To organize the sequence of orders and assignment to waves, consistent with routing, loading and planned departure times of shipping vehicles or production requirement, etc. To reduce space required for shipping deck handling to assemble orders and load; and
2. To assign staff to each wave, with the expectation that all work assigned to each wave providing management with the ability to monitor and manage performance throughout the day, and respond in a timely way to problem that occur and more effectively utilize the staffing throughout the shift.

Warehouse Management includes the improved ability to :-

1. Measure productivity within a function.
2. Budget labour.
3. Estimate the throughout capacity based on staffing levels.
4. Evaluate the impact of changes in methods and equipment by function;
5. Provide feedback regarding performance, and
6. Better understand the nature of the workload as it changes seasonally, as a consequence of demand and as the consequence of sales efforts and marketing campaigns.

For supplier selection we will prepare a “requirement document”, after which we identify and shortlist the suppliers. We will also perform a tender evaluation and selection.

Preliminary Investigations

Due to limited resources and organization can undertake only those projects that are critical to its mission, goals and objectives. Therefore, the goal of preliminary investigations is simply to identify and select a project for development from among all projects that are under construction.

The manual Warehouse Management decreases the efficiency of transaction in terms of time as well as quantity. This management system is unorganized as well as less equipped in terms of resources, thereby being less efficient.

The system being introduced not only overcomes the time barrier in the transactions and inputs but also enhances the workload over the staff, hence making the efficient use of manpower available.

Requirement Planning

There are various requirements for easy flow of information and products within a warehouse

On a software level:-

Operating system – Windows XP, Vista, 7

Database Management -Microsoft Access

On the staff level:-

- Operator
- Workers

User Domain –

- **Operator** – The operator is the person who is responsible for all inputs and analysis of the outputs. He holds the authority to include the outrunned variety in the stock.
- **Worker** – The workers domain is limited to the input of goods to obtain location within the warehouse. They also have the authority to update the tables in the case of picking and putaway at the instant of addition/removal of goods.

Area Requirement – For implementation of WMS estimated area required is 18000 square meter containing 5 sectors of 3000 square meter each.

One maintenance sector, a loading bay and a shipping bay of 1000 sq. Meter each.

Transportation Requirement – 8 heavy duty trucks for distant transportation and shipping and 5 forklifts for transportation within the warehouse.

Coding, Testing & Maintenance

The database is created to include all the storage details with the data type for each value specified, for the items that are in stock in the warehouse, along with the import and export order details. The data in all the tables is linked through the Codes specified to each item. Any change in the import or export order, thus, gets reflected in the storage details. Update queries are used to accordingly update the values and for the calculation of price involved in the transactions of goods.

The database is tested to accept only the data types specified for each value thus affixing an attribute of the entities in the tables. Update queries are also accordingly tested by provision of test values, i.e., dry run.

The tables are maintained as long as the system remains functional, to facilitate addition or deletion of new item varieties in the storage details. It also involves allocation of codes to the new varieties imported and managing the entries when entire stock is to be shipped.